photographs in conjunction with the concept of a moving breakdown front in the bulk of water [108,132-136] provides information on the breakdown dynamics at super-threshold pulse energies.

Through the systematic experimental investigations in a large parameter space, we discovered a new regime of fine-adjustable deterministic energy deposition with temporally smooth UV ns pulses from microchip lasers. This enables material nanoprocessing and precise surgery with cost-effective compact laser sources, which is demonstrated on corneal dissection and refractive index modifications in glass. We explain this finding through the interplay of strong-field ionization (SI), avalanche ionization (AI), recombination and thermal ionization (TI). Here SI provides seed electrons, inhibition of AI by recombination at low electron densities results in nanoeffects, and at higher pulse energies AI combined with TI at larger electron densities leads to a second breakdown step exhibiting the well-known luminescent plasma formation and strong mechanical effects.

Previous work has shown that the plasma energy density in bulk water is limited by the movement of an optical breakdown wave during the laser pulse [108,133,135], and for ns breakdown additionally by an enlargement of the plasma size through radiative energy transport from the regions of primary energy deposition [108]. Consequently, the energy density in plasmas produced by energetic ns laser pulses focused at NA = 0.25 was found to be limited to about 40 kJ/cm^3 [40], which is smaller than in the plasma skin layer produced by ultrashort laser pulses at target surfaces [6,137-139]. In this paper, we determine the average plasma energy density achievable by tight focusing ($0.3 \leq \mathrm{NA} \leq 0.8$) of laser pulses in large (τ_L, λ_L, E_L) parameter space and explore local inhomogeneities and 'hot spots' through high-resolution photography of plasma luminescence. Since regions of primary energy deposition scatter the incoming pump laser light, they can be distinguished from the broadband plasma luminescence. Photographs reveal a fascinating structural complexity, with high-density strings formed during primary energy deposition, which are a consequence of an intrinsic spatial instability of avalanche ionization that is proportional to irradiance *and* local electron density. Energy redistribution by soft X-rays emitted from the strings then inflates the plasma into a larger region with lower average energy density that emits the diffuse luminescence known from previous research.

Our findings pose challenges and provides new impulses for future experimental investigations of the extreme states of matter in laser-induced breakdown as well as for optical breakdown modeling. The spatiotemporal interplay of nonlinear beam propagation and plasma formation in breakdown by ultrashort laser pulses is well covered by existing models [130,131,136,140,141]. However, modeling of the interplay between SI, AI, TI, recombination during nanosecond pulses is still in its infancy, and we

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are not aware of any optical breakdown models covering the intrinsic instability of AI and the role of bremsstrahlung –mediated energy redistribution. Future experimental studies resolving the dynamics of plasma and shock wave formation during ns laser pulses with picosecond temporal and sub-micrometer spatial resolution can provide a quantitative basis for model validation.

The broad experimental coverage of the optical breakdown dynamics in the (τ_L, λ_L, NA, E_L) parameter space provides a solid basis for numerical simulations. However, detailed simulations go beyond the scope of this paper, especially since theoretical models for some aspects of the observed breakdown dynamic still need to be developed. Instead, we present in section 2 a conceptual heuristic framework on the interplay of SI, AI, TI and recombination as a function of irradiance and pulse energy. This framework guides the experiments described in section 3 and is needed to understand the results presented in section 4. In section 5, we first interpret the experimental findings and place them in the context of previous work, and then we use the new insights to derive consequences for laser surgery and material processing in bulk dielectric media, laser ablation of absorbing targets in liquids for nanoparticle generation, fragmentation of the laser-produced nanoparticles, and laser-induced breakdown spectroscopy. Finally, section 6 formulates challenges and a roadmap for future experimental exploration and advanced modelling of plasma formation and plasma-induced hydrodynamic events, which open avenues for novel applications beyond laser surgery, LAL and LIBS. Furthermore, it expands the view towards low-density plasma phenomena below the cavitation bubble threshold and their applications.

Altogether, we establish a comprehensive picture of nonlinear energy deposition, which provides a deeper understanding of the underlying mechanisms, opens new avenues for nanoprocessing with cost-effective compact devices, facilitates parameter selection for material processing and laser surgery, and outlines perspectives for future research in high-energy density physics.

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2. Theoretical considerations and experimental rationale

Optical breakdown is a sequence of nonlinear energy deposition via plasma formation associated with rapid heating and pressure rise in the plasma followed by the emission of an acoustic transient or shock wave and a phase transition with cavitation bubble formation [40,42]. The mechanisms underlying nonlinear energy deposition in the (τ_L, λ_L, E_L) parameter space are very complex – even for focusing at large NA, where nonlinear beam propagation plays little role. In this section, we present an overview of the expected breakdown behavior based on generic equations and heuristic arguments that motivate the experimental rationale and provide a framework for the interpretation of the measurement results.

2.1. Pathways of nonlinear energy deposition and thermalization

The general sequence of nonlinear energy deposition in large-bandgap dielectrics at high irradiance is depicted in **Fig. 1**. Plasma formation is initiated by strong-field ionization (SI) via multiphoton ionization (MPI) or tunneling across the band gap between valence band (VB) and conduction band (CB). Seed electron generation is followed by avalanche ionization (AI) that consists of a sequence of inverse bremsstrahlung absorption (IBA) events of single photons through which the kinetic energy of the CB electrons is raised until it suffices to free another VB electron by impact ionization. Conservation of energy and momentum during IBA requires that the photon interacts with the electron during a collision with a heavy particle (atom or ion). Moreover, it requires a kinetic energy of the impacting electron $\geq (3/2)\,\tilde{\Delta}$ [37,142,143], where $\tilde{\Delta}$ is the ionization potential of the dielectric. This potential is given by the sum of band gap energy E_{gap} and the oscillation (quiver) energy E_{osc} of CB electrons in the strong electromagnetic field [37,144]. Avalanche ionization initiated by strong-field ionization is the backbone of nonlinear energy deposition into large band gap materials such as water because it leads to a rapid multiplication of the number of free electrons [6,36,37,143].

The energy deposited into the electronic system is thermalized via collisional electron-phonon coupling and electron-hole recombination. The resulting volumetric energy density U is given by the time-integrated product of CB electron density, ρ_c, and the average energy ε_{avg} carried by CB electrons, where ε_{avg} is the sum of $\tilde{\Delta}$ and the average kinetic energy of the electrons, $\varepsilon_{kin.avg}$. For fs breakdown, only one set of free electrons is produced during the laser pulse, and their energy is thermalized afterwards [22,37,115]. Here, U is simply the product ($\rho_c \times \varepsilon_{avg}$) but for ns breakdown, it can be much larger. When a sufficiently high thermodynamic temperature, T_{equil}, of the breakdown region is reached, VB electrons can be lifted into the CB via thermal ionization. For sufficiently long pulse durations,

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